

Re-normalization and the functional form

Lessons from Agostinelli and Wiswall (2025)

강현제

중앙대학교 경제학과 BK21 사업팀 특강
교토대학교 경제연구소

2026년 8월 31일–9월 2일

Where We Are Going

1. Main idea 1: CES and its implications on the dynamic skill formation
2. Main idea 2: we care about growth or decline of skills
3. Measurement restriction should be age-invariant
4. An application using CNLSY79

Three Policy Questions

1. When is investment most productive?

$$\frac{\partial \theta_T}{\partial I_t}$$

어느 시점의 투자가 최종 능력에 가장 큰 영향을 미치는가?

2. Who benefits most from investment?

$$\frac{\partial^2 \theta_{t+1}}{\partial \theta_t \partial I_t}$$

능력 수준에 따라 투자의 효과가 달라지는가?

3. How do early and later investments interact?

$$\frac{\partial^2 \theta_T}{\partial I_t \partial I_{t+1}}$$

초기 투자가 후속 투자의 생산성을 높이는가?

The Dynamic Skill-Formation Model

For a child i , let θ_{it} denote the stock of skills and l_{it} the flow of parental investment.

$$\theta_{i,t+1} = f(\theta_{it}, l_{it}, \eta_{it}), \quad t = 1, 2, \dots, T.$$

- ▶ θ_{it} is a state variable: it carries information from the past.
- ▶ l_{it} is a current input chosen by the family.
- ▶ η_{it} collects shocks and omitted inputs.
- ▶ The technology can change with developmental stage t .

The CHS Benchmark

The CES technology used in much of the earlier literature can be written as

$$\theta_{t+1} = [\gamma_{1,t}\theta_t^{\varphi_t} + \gamma_{2,t}l_t^{\varphi_t}]^{1/\varphi_t}.$$

With positive input weights and $\varphi_t \leq 1$,

$$\frac{\partial^2 \theta_{t+1}}{\partial \theta_t \partial l_t} \geq 0.$$

- ▶ Current skill and current investment are weak complements.
- ▶ Investment is mechanically more productive for higher-skilled children.
- ▶ However, this does not rule out early intervention being optimal through dynamic effects.

AW's Main Idea 1: Trans-log

Agostinelli–Wiswall use a flexible translog specification:

$$\ln \theta_{t+1} = \ln A_t + \gamma_{1,t} \ln \theta_t + \gamma_{2,t} \ln l_t + \gamma_{3,t} \ln l_t \ln \theta_t + \eta_{\theta,t}$$

The investment elasticity is

$$e_t^l(\theta_t) \equiv \frac{\partial \ln \theta_{t+1}}{\partial \ln l_t} = \gamma_{2,t} + \gamma_{3,t} \ln \theta_t.$$

- ▶ $\gamma_{1,t}$: self-productivity in logs
- ▶ $\gamma_{2,t}$: investment elasticity at $\ln \theta_t = 0$
- ▶ $\gamma_{3,t}$: heterogeneity in investment productivity

What Does the Interaction Coefficient Mean?

$$\frac{\partial e_t^l(\theta_t)}{\partial \ln \theta_t} = \frac{\partial^2 \ln \theta_{t+1}}{\partial \ln \theta_t \partial \ln l_t} = \gamma_{3,t}.$$

Interpretation

- ▶ $\gamma_{3,t} > 0$: direct/log complementarity; investment is more productive for high-skill children.
- ▶ $\gamma_{3,t} < 0$: compensatory investment; investment is more productive for low-skill children.
- ▶ $\gamma_{3,t} = 0$: the investment elasticity is homogeneous in skill.

Static and Dynamic Complementarity

Static / direct

Does current skill change the productivity of current investment?

$$\frac{\partial^2 \ln \theta_{t+1}}{\partial \ln \theta_t \partial \ln I_t} = \gamma_{3,t}$$

Dynamic

Does early investment change the productivity of later investment through the skill stock it creates?

$$\frac{\partial}{\partial I_t} \left(\frac{\partial \theta_T}{\partial I_{t+1}} \right)$$

- ▶ Early investment can still raise later skills and later investment returns.
- ▶ The two concepts answer different policy questions.

The Dynamic Effect of Early Investment

The effect of early investment on final skill is

$$\begin{aligned}\frac{\partial \ln \theta_{t+2}}{\partial \ln l_t} &= \frac{\partial \ln \theta_{t+2}}{\partial \ln \theta_{t+1}} \frac{\partial \ln \theta_{t+1}}{\partial \ln l_t} \\ &= \underbrace{(\gamma_{1,t+1} + \gamma_{3,t+1} \ln l_{t+1})}_{\text{skill persistence}} \underbrace{(\gamma_{2,t} + \gamma_{3,t} \ln \theta_t)}_{e_t^l(\theta_t)}.\end{aligned}$$

Therefore,

$$\boxed{\frac{\partial \ln \theta_{t+2}}{\partial \ln l_t} = e_{t+1}^\theta(l_{t+1}) e_t^l(\theta_t)}$$

Even if $\gamma_{3,t} < 0$, early investment can still increase final skill when both components are positive.

Dynamic Complementarity Across Investment Dates

On the log scale,

$$\begin{aligned}\frac{\partial^2 \ln \theta_{t+2}}{\partial \ln l_t \partial \ln l_{t+1}} &= \frac{\partial}{\partial \ln l_{t+1}} \left[e_{t+1}^\theta(l_{t+1}) e_t^l(\theta_t) \right] \\ &= \boxed{\gamma_{3,t+1} e_t^l(\theta_t)}.\end{aligned}$$

If $e_t^l(\theta_t) > 0$, then

$$\text{sign} \left(\frac{\partial^2 \ln \theta_{t+2}}{\partial \ln l_t \partial \ln l_{t+1}} \right) = \text{sign}(\gamma_{3,t+1}).$$

Key point

The first-period interaction $\gamma_{3,t}$ does not determine dynamic complementarity between l_t and l_{t+1} .

The relevant interaction is $\gamma_{3,t+1}$, because it determines how skill at $t+1$ changes the productivity of investment at $t+1$.

AW's Main Idea 2

Normalize λ and μ to "a" period t

- ▶ Test scores have arbitrary locations and scales
- ▶ A test may become more sensitive to a latent skill as children age
- ▶ If the measurement scale changes with age, apparent skill growth can be confused with a change in the technology.
- ▶ Therefore, age-by-age normalization can change the object being compared.

(Review) We Observe Measures, Not Skills

A test score is a noisy measure of a latent skill.

$$Z_{t,m} = \mu_{t,m} + \lambda_{t,m} \ln \theta_t + \epsilon_{t,m}, \quad m = 1, \dots, M_t.$$

- ▶ $Z_{t,m}$: observed test or assessment
- ▶ $\mu_{t,m}$: location of the measure
- ▶ $\lambda_{t,m}$: scale / factor loading
- ▶ $\epsilon_{t,m}$: measure-specific noise

Initial-Period Measurement: Three Measures

Suppose three measures are available at the initial period:

$$Z_{0,m} = \mu_{0,m} + \lambda_{0,m} \ln \theta_0 + \epsilon_{0,m}, \quad m = 1, 2, 3.$$

Assume mutually independent measurement errors and normalize $\lambda_{0,1} = 1$.

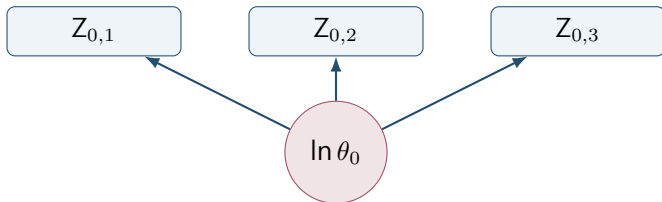
$$C_{12} = \lambda_{0,2} V_0, \quad C_{13} = \lambda_{0,3} V_0, \quad C_{23} = \lambda_{0,2} \lambda_{0,3} V_0,$$

where $V_0 = \text{Var}(\ln \theta_0)$.

$$V_0 = \frac{C_{12} C_{13}}{C_{23}}$$

$$\lambda_{0,2} = \frac{C_{12}}{V_0}, \quad \lambda_{0,3} = \frac{C_{13}}{V_0}$$

What the Covariances Are Doing



common covariance identifies the latent signal

- ▶ C_{12} , C_{13} and C_{23} share the same latent variance.
- ▶ Their ratios recover relative factor loadings.
- ▶ The idiosyncratic parts $\epsilon_{0,m}$ do not generate cross-measure covariance.

The Problem Begins After Period 1

Without further restrictions, the later-period measurement equation is

$$Z_{t,m} = \mu_{t,m} + \lambda_{t,m} \ln \theta_t + \epsilon_{t,m}.$$

Even if later-period covariances identify the factor up to a scale, that scale need not be the same as at period 1.

$$\widetilde{\ln \theta_t} = a_t + b_t \ln \theta_t, \quad b_t > 0.$$

- ▶ Each t can have its own location a_t and scale b_t .
- ▶ The production function and the measurement equation can absorb this transformation.
- ▶ Therefore, latent skill growth and technology parameters are not separately determined.

시점별로 factor loading을 다시 1로 만드는 것은 단순한 단위 선택이 아니라, 시점 간 비교에 제약을 추가할 수 있음

Why Re-normalization Changes the Question

Consider the observed mean change:

$$\begin{aligned} E(Z_{t+1,m}) - E(Z_{t,m}) &= (\mu_{t+1,m} - \mu_{t,m}) \\ &\quad + (\lambda_{t+1,m} - \lambda_{t,m})E(\ln \theta_t) \\ &\quad + \lambda_{t+1,m} [E(\ln \theta_{t+1}) - E(\ln \theta_t)] . \end{aligned}$$

The observed change can reflect:

1. a change in the mean of the measure;
2. a change in the sensitivity of the measure to skill;
3. genuine growth in latent skill.

$\lambda_{t+1,m}$ 이 Key Parameter. 매 t마다 normalization하면
 $\lambda_{t+1,m}$ 의 식별이 혼란스러워짐

The Same Problem for Investment Productivity

The observed sensitivity of a measure to investment is

$$\frac{\partial E(Z_{t+1,m} \mid I_t)}{\partial I_t} = \lambda_{t+1,m} \frac{\partial E(\ln \theta_{t+1} \mid I_t)}{\partial I_t}.$$

Comparing two ages gives

$$\begin{aligned} & \frac{\partial E(Z_{t+1,m} \mid I_t)}{\partial I_t} - \frac{\partial E(Z_{t,m} \mid I_{t-1})}{\partial I_{t-1}} \\ &= \lambda_{t+1,m} \frac{\partial E(\ln \theta_{t+1} \mid I_t)}{\partial I_t} - \lambda_{t,m} \frac{\partial E(\ln \theta_t \mid I_{t-1})}{\partial I_{t-1}}. \end{aligned}$$

- ▶ A changing observed sensitivity need not imply a changing primitive return.
- ▶ This is why “critical periods” require cross-age measurement restrictions.

Age-Invariance Example 1: Vocab

Suppose Z_t^V measures a child's vocabulary skill.

The intuition

A useful measure should act like the same ruler at different ages.

$$\theta_5^V = \theta_{10}^V \implies E[Z_5^V | \theta_5^V] = E[Z_{10}^V | \theta_{10}^V].$$

Consider a vocabulary test with words of increasing difficulty:

Z_t^V = number of words correctly defined.

- ▶ A five-year-old and a ten-year-old with the same latent vocabulary skill should obtain the same expected score.
- ▶ A ten-year-old will normally score higher because $\theta_{10}^V > \theta_5^V$, not because the measurement rule changed.
- ▶ This allows observed score changes to be interpreted as latent skill growth.

The Ideal Case: A Perfect Skill Meter

Imagine a hypothetical test that measures mathematical skill perfectly.

For item j ,

$$\Pr(R_{itj} = 1 \mid \theta_{it}^M, a_{it}, b_j) = \Lambda(\theta_{it}^M - b_j),$$

where b_j is the item difficulty.

The probability of answering an item correctly depends only on latent skill and item difficulty—not directly on age.

$$\Pr(R_{itj} = 1 \mid \theta_{it}^M, a_{it} = 5, b_j) = \Pr(R_{itj} = 1 \mid \theta_{it}^M, a_{it} = 10, b_j).$$

Therefore,

$$E[Z_t^M \mid \theta_t^M = \theta] = g(\theta) \quad \text{for every age } t.$$

Why this is almost impossible

In reality, age may affect language, test-taking strategies, attention, fatigue, schooling exposure, and familiarity with the items.

Thus, age invariance is an approximation that must be defended for each measure—not a property that can be created automatically by standardizing scores.

Age-Invariant Does Not Mean Age-Standardized

- ▶ It does not require that all children answer the same questions.
- ▶ It does not mean subtracting the age-specific mean and dividing by the age-specific standard deviation.
- ▶ It does not require the skill distributions to overlap across ages.
- ▶ It means that the measurement relationship is stable for a common latent skill level.

Example

PIAT Mathematics is designed to measure achievement over a broad age range. AW argue that its scale can plausibly be treated as age-invariant from ages 5–14.

중요한 것은 모든 측정치가 age-invariant일 필요는 없다는 점이다. 적어도 하나의 적절한 연결고리가 있으면 된다.

Age-Invariant Measures

Definition

A pair of measures $Z_{t,m}$ and $Z_{t+1,m}$ is age-invariant if

$$E(Z_{t,m} \mid \theta_t = p) = E(Z_{t+1,m} \mid \theta_{t+1} = p) \quad \text{for all } p.$$

Under the linear measurement model, this implies

$$\mu_{t,m} = \mu_{t+1,m} = \mu_m, \quad \lambda_{t,m} = \lambda_{t+1,m} = \lambda_m.$$

- ▶ The same latent skill level has the same expected score at different ages.
- ▶ The measurement scale is linked across periods.
- ▶ The production technology can vary with age without being confounded with the scale.

Under Age-Invariance: A Common Ruler

For an age-invariant measure,

$$Z_{t,m} = \mu_m + \lambda_m \theta_t + \epsilon_{t,m},$$

and

$$Z_{t+1,m} = \mu_m + \lambda_m \theta_{t+1} + \epsilon_{t+1,m}.$$

Subtracting the two equations,

$$Z_{t+1,m} - Z_{t,m} = \lambda_m (\theta_{t+1} - \theta_t) + (\epsilon_{t+1,m} - \epsilon_{t,m}).$$

Therefore, under mean-independent measurement error,

$$E[Z_{t+1,m} - Z_{t,m} \mid \theta_t, \theta_{t+1}] = \lambda_m (\theta_{t+1} - \theta_t).$$

Interpretation

The same measure provides a common ruler across ages.

After fixing the initial scale, changes in the measure can be linked to latent skill growth.

Only One Measure Needs to Be Age-Invariant

Let m be an age-invariant measure:

$$Z_{t,m} = \mu_m + \lambda_m \theta_t + \epsilon_{t,m}.$$

Other measures j may be age-specific:

$$Z_{t,j} = \mu_{t,j} + \lambda_{t,j} \theta_t + \epsilon_{t,j}.$$

If measurement errors are uncorrelated,

$$\text{Cov}(Z_{t,j}, Z_{t,m}) = \lambda_{t,j} \lambda_m \text{Var}(\theta_t).$$

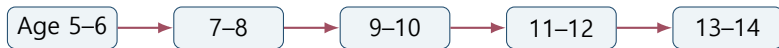
- ▶ The age-invariant measure m links the latent skill scale across ages.
- ▶ Covariances with other measures recover their common latent component.
- ▶ The remaining measures need not themselves be age-invariant.
- ▶ Multiple measures are still needed to separate signal from measurement noise.

Age-Invariance in the Skill-Formation Literature

Paper	Age-invariance	Normalization	Cross-age link
Cunha–Heckman (2007)	N/A: theory	Recursive skill-formation technology	Measurement system and age-invariance are not separately formulated.
Cunha–Heckman–Schennach (2010)	Not explicitly imposed	for every t	The factor scale is normalized separately at each period. Adult outcomes, especially schooling, anchor final skills.
Attanasio–Meghir–Nix–Salvati (2017)	Not explicitly imposed	Time-varying semi-log measurement:	Uses CHS-style period-specific normalization. No common age-invariant measure is imposed.
Attanasio–Meghir–Nix (2020)	Yes: selected anchors	Initial-period location normalization and common anchor measures across periods.	PPVT for cognition and height-for-age z-score for health. Other measures need not be age-invariant.
Agostinelli–Wiswall (2016/2025)	Yes: at least one per skill	Initial-period scale and location normalization. Later factor scales are linked through age-invariant measures.	PIAT Mathematics is the main cognitive anchor in the CNLSY79 application. Not every measure must be invariant.

AW Application: Data and Developmental Stages

- ▶ **Data:** Children and Young Adults of the NLSY79 (CNLSY79)
- ▶ **Sample:** 11,509 children in the main application
- ▶ **Initial period:** ages 5–6
- ▶ **Final childhood period:** ages 13–14
- ▶ Four two-year transitions: 5–6 → 7–8 → 9–10 → 11–12 → 13–14



The data follow children from age 5–6 through adulthood using the NLSY79 Child and Young Adult surveys.

What Is Measured in the Application?

Latent object	Observed measures	Role
Child cognitive skill	PIAT Mathematics, Reading Recognition, Reading Comprehension, PPVT	Multiple noisy measures; PIAT Mathematics is the age-invariant anchor.
Parental investment	HOME score items, reading, praise, museum visits, and related measures	Latent quantity and quality of parental investment.
Mother's cognitive skill	ASVAB subscales	Initial condition and predictor of investment.
Mother's noncognitive skill	Rotter and Rosenberg indexes	Initial condition and predictor of investment.
Adult outcomes	Highest grade completed at age 23 or older; earnings at age 29	Economically interpretable validation and policy metrics.

측정모형은 아동의 능력뿐 아니라 투자와 어머니의 능력도 latent하게 처리한다.

The Age-Invariant Anchor in Practice

For PIAT Mathematics, AW impose a common measurement relationship:

$$Z_{t,\text{Math}} = \mu_{\text{Math}} + \lambda_{\text{Math}} \ln \theta_t + \epsilon_{t,\text{Math}}, \quad t = 0, 1, 2, 3, 4.$$

The remaining measures help recover the latent factor and measurement error:

$$Z_{t,m} = \mu_{t,m} + \lambda_{t,m} \ln \theta_t + \epsilon_{t,m}.$$

- ▶ PIAT Math links the skill unit across ages.
- ▶ Recognition and Comprehension provide additional information and instruments.
- ▶ The measurement relationship is an assumption about the test, not a claim that children stop developing.

Is PIAT Mathematics Age-Invariant?

The required restriction

For PIAT Mathematics to link cognitive skill across ages,

$$E(Z_{t,PIAT} \mid \theta_{C,t} = p) = E(Z_{t+1,PIAT} \mid \theta_{C,t+1} = p)$$

must hold for all skill levels p .

Under a linear measurement model, this means

$$Z_{t,PIAT} = \mu_{PIAT} + \lambda_{PIAT}\theta_{C,t} + \epsilon_{t,PIAT}, \quad \text{for every } t.$$

Why is PIAT Mathematics a plausible candidate?

- ▶ It is designed to measure the same broad cognitive construct over a relatively wide age range.
- ▶ The test provides a vertically linked scale, allowing scores from different ages to be compared.
- ▶ The same measure is observed repeatedly in the CNLSY79.
- ▶ It therefore provides a natural bridge for the latent cognitive skill scale across childhood periods.

PIAT (Cont'd)

But age-invariance is not automatic.

- ▶ Test content, ceiling effects, floor effects, and practice effects may change the mapping from skill to observed score.
- ▶ Age-standardizing a score does not by itself establish age-invariance.
- ▶ The restriction should be interpreted as a maintained identifying assumption, supported by the design of the PIAT and the factor model.

Interpretation

PIAT Mathematics is not necessarily perfectly age-invariant. It is a particularly plausible **bridge measure** for connecting cognitive skill across ages.

A Signal-and-Noise Reading of the Measures

For a measure $Z_{t,m}$, the signal share is

$$\text{SignalShare}_{t,m} = \frac{\lambda_{t,m}^2 \text{Var}(\ln \theta_t)}{\lambda_{t,m}^2 \text{Var}(\ln \theta_t) + \text{Var}(\epsilon_{t,m})}.$$

Age	Measure	Signal share
5–6	PIAT Mathematics	0.273
7–8	PIAT Mathematics	0.764
9–10	PIAT Mathematics	0.777
11–12	PIAT Mathematics	0.804
13–14	PIAT Mathematics	0.924

- ▶ At the initial age, the normalized Math measure contains substantial noise.
- ▶ The same measure becomes more informative at later ages.
- ▶ This is exactly why the loading must be linked across ages rather than reset mechanically.

Signal shares are reported from the measurement estimates in the AW appendix.

Normalization Changes the Skill Unit

Let $\ell_t \equiv \ln \theta_t$.

Cross-age normalization: AW

$$E(\ell_0) = 0, \quad \lambda_{0,m^*} = 1,$$

and for an age-invariant measure m^* ,

$$\mu_{t,m^*} = \mu_{0,m^*}, \quad \lambda_{t,m^*} = \lambda_{0,m^*} \quad \text{for every } t.$$

$$\tilde{\ell}_t = a_t + b_t \ell_t$$

If b_t varies with age, changes in observed scores can be confused with changes in the skill-formation technology.

Investment Function: Kept in the Background

AW model parental investment as

$$\ln I_t = \alpha_{1,t} \ln \theta_t + \alpha_{2,t} \ln \theta_{MC} + \alpha_{3,t} \ln \theta_{MN} + \alpha_{4,t} \ln Y_t + \eta_{I,t}.$$

- ▶ Investment responds to the child's skills, maternal skills, and income.
- ▶ $\alpha_{1,t} > 0$ is reinforcement; $\alpha_{1,t} < 0$ is compensation in the investment choice equation.
- ▶ This is a reduced-form approximation to parental behavior, not a complete household-choice model.
- ▶ The investment equation matters because I_t is endogenous and measured with error.

Estimate the Skill Technology with IV

Construct a residual measure of the latent variable:

$$\tilde{Z}_{\theta,t,m} \equiv \frac{Z_{\theta,t,m} - \mu_{t,m}}{\lambda_{t,m}} = \ln \theta_t + e_{\theta,t,m}.$$

Substituting noisy measures into the translog technology gives a reduced form:

$$\begin{aligned} Z_{\theta,t+1,m} = & \delta_{0,t,m} + \delta_{1,t,m} \tilde{Z}_{\theta,t,m'} + \delta_{2,t,m} \tilde{Z}_{l,t,m'} \\ & + \delta_{3,t,m} \tilde{Z}_{\theta,t,m'} \tilde{Z}_{l,t,m'} + \pi_{t,m}. \end{aligned}$$

$$\gamma_{j,t} = \frac{\delta_{j,t,m}}{\lambda_{\theta,t+1,m}}, \quad j = 1, 2, 3$$

What Identifies the Interaction?

The interaction coefficient is identified from variation in

$$\tilde{Z}_{\theta,t,m'} \times \tilde{Z}_{l,t,m'}.$$

The excluded instruments include

$$\tilde{Z}_{\theta,t,m''}, \quad \tilde{Z}_{l,t,m''}, \quad \tilde{Z}_{\theta,t,m''} \times \tilde{Z}_{l,t,m''}.$$

- ▶ The alternative measures share the same latent variables.
- ▶ Their measurement errors are assumed independent across measures.
- ▶ The factor loadings must already be known or identified.
- ▶ The age-invariant restriction supplies the cross-age scale needed for $\lambda_{t+1,m}$.

Measurement restrictions make a flexible technology estimable.

AW Estimates: Skill Technology

	5–6	7–8	9–10	11–12
Log skills $\gamma_{1,t}$	1.966 (0.153)	1.086 (0.035)	0.897 (0.027)	1.065 (0.029)
Log investment $\gamma_{2,t}$	0.799 (0.261)	0.695 (0.337)	0.713 (0.403)	0.252 (0.538)
Log skills $\times$ log investment $\gamma_{3,t}$	−0.105 (0.066)	−0.005 (0.019)	−0.003 (0.013)	0.003 (0.010)
Return to scale (at sample means)	2.660	1.776	1.606	1.320
Standard deviation of shocks	5.612	4.519	3.585	4.019

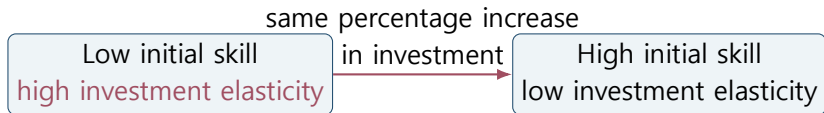
- ▶ Skills are self-productive and investment raises next-period skill.
- ▶ The early interaction estimate is negative: the investment elasticity falls with current skill.
- ▶ The interaction is close to zero in later periods.

Estimates from AW Table 2; standard errors are in parentheses.

Reading the Early Interaction Coefficient

At ages 5–6,

$$e_0^l(\theta_0) = 0.799 - 0.105 \ln \theta_0.$$



- ▶ For low-skill children, the investment elasticity approaches about 1.5.
- ▶ For high-skill children, it approaches about 0.2.
- ▶ This is compensatory static complementarity, not a claim that investment is useless for high-skill children.

Early Investment and Skill Persistence

The estimated technology implies that early inputs matter through two channels:

$$l_0 \longrightarrow \theta_1 \longrightarrow \theta_2 \longrightarrow \theta_3 \longrightarrow \theta_4.$$

- ▶ The age 5–6 production function has high self-productivity and high investment productivity.
- ▶ The estimated total factor productivity falls sharply by ages 11–12.
- ▶ Early investment can therefore have a large effect on final skills even if later investment remains important.

Careful interpretation

The age-invariant measure is what allows AW to compare these coefficients across stages. Without it, a larger early coefficient could partly reflect a different measurement scale.

Adult Outcomes as an Economic Metric

AW link final latent skills to adult outcomes:

$$S_{23} = \mu_S + \alpha_S \ln \theta_4 + \eta_S,$$

$$\ln W_{29} = \mu_W + \alpha_W \ln \theta_4 + \eta_W.$$

Their estimates imply approximately

$$\frac{\partial S_{23}}{\partial \ln \theta_4} \approx 0.15 \text{ years of schooling}, \quad \frac{\partial \ln W_{29}}{\partial \ln \theta_4} \approx 0.041.$$

- ▶ Skill units are still constructed, but their consequences are economically interpretable.
- ▶ The adult equations help translate skill changes into policy outcomes.
- ▶ Schooling is observed with less censoring than realized earnings.

Policy Experiment: A \$10,000 Income Transfer

		5–6	7–8	9–10	11–12
Measurement-error	cor-	0.300	0.172	0.129	0.050
	rected				
Not corrected		0.062	0.029	0.022	0.013

- ▶ Entries are additional months of completed schooling at age 23.
- ▶ The same transfer has the largest estimated effect when given at age 5–6.
- ▶ Ignoring measurement error reduces the estimated age 5–6 effect by almost a factor of five.

AW Table 3, Panel A. The transfer is simulated through the estimated investment and skill-formation system.

Who Benefits from the Transfer?

Low-income families

- ▶ The age 5–6 transfer raises completed schooling by about one month or more for the lowest income percentiles.
- ▶ The effect is close to zero for the richest families.

Low-skill children

- ▶ The transfer effect is also larger for children with low initial skills.
- ▶ The heterogeneity is smaller than the heterogeneity by family income.

Targeting by both income and initial skill is most informative.

What Does AW Change Relative to CHS?

1. Static complementarity

$$\gamma_{3,0} < 0$$

makes investment more productive for low-skill children at age 5–6.

2. Timing

early investment productivity > later investment productivity

after accounting for the common measurement scale.

3. Measurement error

corrected policy effect $\gg$ uncorrected policy effect.

Flexible technology plus explicit measurement restrictions
changes the policy conclusions.

What Remains an Assumption?

- ▶ The log-linear measurement model imposes a cardinal structure on observed measures.
- ▶ At least one measure, especially PIAT Mathematics, is treated as age-invariant.
- ▶ Measurement errors are independent across measures and independent of latent variables in the required moments.
- ▶ The investment equation is a reduced-form approximation to household behavior.
- ▶ Counterfactuals require additional assumptions about distributions and dynamic shocks.

Application: Genes and Skill Formation

Houmark, Ronda, and Rosholm (2024)

Central question

How do genetic endowments and parental investments interact in the formation of children's skills?

- ▶ Are genes and family investments substitutes or complements?
- ▶ Do parents respond to children's genetic endowments?
- ▶ How much of the genetic association with skills operates through parental investments?
- ▶ Can public investment reduce skill inequality even when genes matter?

Nature and nurture are not separate determinants of skill formation.

Three Genetic Mechanisms

The paper distinguishes three channels through which genes may affect children's skill development.

Child genes $\longrightarrow$ Child skills

Direct genetic effect

Child genes $\longrightarrow$ Parental investments $\longrightarrow$ Child skills

Nurture of nature

Parental genes $\longrightarrow$ Parental investments $\longrightarrow$ Child skills

Nature of nurture

- ▶ Child genes may affect the child's ability to learn from investments.
- ▶ Child genes may change how parents respond to the child.
- ▶ Parental genes may affect parenting quality, resources, and preferences.

Data: ALSPAC

Avon Longitudinal Study of Parents and Children

- ▶ Longitudinal British data covering early childhood.
- ▶ Children are observed from approximately age 0 to age 7.
- ▶ Child skills and parental investments are observed repeatedly.
- ▶ Molecular genetic information is available for children and both parents.

$$t \in \{[0, 2), [2, 3), [3, 4), [4, 5), [5, 6), [6, 7)\}.$$

The genetic endowment is measured using an educational-attainment polygenic score:

$$G_i^C = \text{child's EA polygenic score}, \quad G_i^P = \text{parental EA polygenic score}.$$

The empirical analysis focuses on mechanisms during early childhood.

Initial Skills and Genetic Endowments

Initial skills are modeled as a function of child and parental genetic endowments:

$$\ln \theta_{i0} = \alpha_1 G_i^C + \alpha_2 G_i^P + \alpha_X X_{i0} + \varepsilon_{i0}.$$

- ▶ G_i^C : direct genetic contribution to initial skills.
- ▶ G_i^P : parental genetic contribution to initial skills.
- ▶ X_{i0} : gender, birth order, and other predetermined characteristics.

Important limitation

Initial skills are measured at ages 0–2, but parental investments during this very early period are not observed.

Parental genes may also capture prenatal conditions and unobserved early investment

Genetic Endowments in the Investment Function

Parental investments depend on the child's current skill, the child's genes, and the parents' genes:

$$\ln I_{it} = \gamma_{1,t} \ln \theta_{it} + \gamma_{2,t} G_i^C + \gamma_{3,t} G_i^P + \gamma_{X,t} X_{it} + \eta_{it}.$$

- ▶ $\gamma_{1,t}$: parental response to the child's current skill.
- ▶ $\gamma_{2,t}$: parental response to the child's genes, conditional on current skill and parental genes.
- ▶ $\gamma_{3,t}$: effect of parental genes on parental investments.

$\gamma_{2,t} \Rightarrow$ nurture of nature

$\gamma_{3,t} \Rightarrow$ nature of nurture

Measurement System

Skills and investments are latent variables measured with error.

$$m_{ijt}^{\theta} = \mu_{jt}^{\theta} + \lambda_{jt}^{\theta} \ln \theta_{it} + \nu_{ijt}^{\theta},$$

$$m_{ikt}^l = \mu_{kt}^l + \lambda_{kt}^l \ln l_{it} + \nu_{ikt}^l.$$

- ▶ Several observed measures are used for each latent factor.
- ▶ Measurement errors are assumed to be independent across measures and developmental periods.
- ▶ Covariances across measures identify factor loadings and latent-factor variance.
- ▶ The method follows the latent-factor structure of CHS and the measurement strategy of AW.

Why this matters

A single test score or a single investment measure would combine true variation with measurement noise.

Age Scale and Overlapping Measures

There is no single child-skill measure observed at every age.

For parental investments, the paper uses a measure observed throughout the relevant periods:

$$m_{i1t}^l = 0 + 1 \cdot \ln l_{it} + \nu_{i1t}^l.$$

For child skills, two partially overlapping measures are used:

$$m_{i1t}^\theta = 0 + 1 \cdot \ln \theta_{it} + \nu_{i1t}^\theta, \quad t \in \{0, 1, 2\},$$

$$m_{i2t}^\theta = \mu_{21} + \lambda_{21} \ln \theta_{it} + \nu_{i2t}^\theta, \quad t \in \{1, 2, 3, 4, 5\}.$$

- ▶ "Can build a tower of 8 bricks": early periods.
- ▶ "Can play card games": later periods.
- ▶ The overlap at periods 1 and 2 links the latent-skill scale.

Estimation: A Modified AW Algorithm

The estimation proceeds sequentially.

1. Estimate the measurement system using covariance restrictions.
2. Construct residual measures of latent skills and investments:

$$\tilde{m}_{ijt}^{\theta} = \frac{m_{ijt}^{\theta} - \hat{\mu}_{jt}^{\theta}}{\hat{\lambda}_{jt}^{\theta}} = \ln \theta_{it} + \text{measurement error}.$$

3. Estimate the investment and skill-formation equations.
4. Use alternative measures as instruments for the residual measures that contain measurement error.

Important distinction

The alternative-measure IV corrects measurement error. The genetic variation is used to identify genetic channels. These are two separate sources of identification.

Main Parameter Estimates

	$\ln \theta_{i0}$	$\ln \theta_{i,t+1}$	$\ln l_{it}$
Child's EA PGS	0.022 [0.002, 0.039]	0.016 [0.005, 0.032]	0.013 [-0.001, 0.027]
Parental EA PGS	-0.001 [-0.018, 0.018]	0.020 [0.010, 0.037]	0.041 [0.023, 0.056]
$\ln \theta_{it}$	—	0.469 [0.419, 0.538]	0.265 [0.180, 0.303]
$\ln l_{it}$	—	0.205 [0.120, 0.293]	—
Constant	1.463 [1.434, 1.494]	1.151 [0.672, 1.567]	3.076 [2.985, 3.295]

- ▶ The self-productivity coefficient is 0.469.
- ▶ The investment coefficient in the skill-formation equation is 0.205.
- ▶ Parental genes strongly predict parental investment.
- ▶ The direct effect of the child's genes on investment is small and imprecise.

Brackets contain 90% bootstrap confidence intervals.

Variance Decomposition by Age

Panel A: Child skills

	Age					
	0-2	2-3	3-4	4-5	5-6	6-7
Child's EA PGS	1.55%	4.07%	5.65%	6.63%	7.11%	7.39%
Parental EA PGS	-0.00%	1.49%	3.08%	4.15%	4.70%	4.96%
Observed characteristics	6.81%	1.79%	0.99%	0.71%	0.58%	0.51%
Unobserved factors	91.64%	92.66%	90.29%	88.51%	87.61%	87.14%

Panel B: Parental investments

	Age				
	0-2	2-3	3-4	4-5	5-6
Child's EA PGS	4.06%	5.01%	5.61%	5.87%	6.06%
Parental EA PGS	4.85%	6.30%	7.21%	7.61%	7.95%
Observed characteristics	6.97%	5.79%	5.26%	5.07%	4.93%
Unobserved factors	84.12%	82.90%	81.92%	81.45%	81.06%

The table decomposes the variance of latent skills and investments into genetic, observed, and unobserved components.

Mechanisms Decomposition by Age

	Age					
	0–2	2–3	3–4	4–5	5–6	6–7
Panel A: Child's skills						
Nurture of nature	0.00%	5.91%	7.74%	9.12%	10.20%	10.90%
Nature of nurture	-3.75%	41.61%	47.34%	48.73%	49.29%	49.43%
Direct effect	103.75%	52.48%	44.92%	42.15%	40.51%	39.67%
Panel B: Parental investments						
Nurture of nature	33.99%	41.12%	44.18%	46.19%	46.74%	—
Nature of nurture	66.01%	58.88%	55.82%	53.81%	53.26%	—

- ▶ The direct genetic component becomes less dominant over time.
- ▶ The nature-of-nurture channel becomes increasingly important for the skill–gene association.
- ▶ Parental genes explain a substantial part of the genetic association with parental investments.

The decomposition is model-based; components can be negative or exceed 100% because the channels are defined relative to counterfactual changes.

How Should We Read the Results?

1. Persistence is not purely genetic

The coefficient on current skill is

$$\hat{\delta}_1 = 0.469.$$

This captures the persistence of the existing skill stock, conditional on measured genetic endowments and investment.

How Should We Read the Results?

1. Persistence is not purely genetic

The coefficient on current skill is

$$\hat{\delta}_1 = 0.469.$$

This captures the persistence of the existing skill stock, conditional on measured genetic endowments and investment.

2. Genes shape the investment environment

Parental genes strongly predict parental investment:

$$\hat{\gamma}_{\text{parental PGS}} = 0.041.$$

Thus, genetic differences are associated not only with children's learning capacity, but also with the investment environment they receive.

How Should We Read the Results?

1. Persistence is not purely genetic

The coefficient on current skill is

$$\hat{\delta}_1 = 0.469.$$

This captures the persistence of the existing skill stock, conditional on measured genetic endowments and investment.

2. Genes shape the investment environment

Parental genes strongly predict parental investment:

$$\hat{\gamma}_{\text{parental PGS}} = 0.041.$$

Thus, genetic differences are associated not only with children's learning capacity, but also with the investment environment they receive.

3. The genetic association grows dynamically

The model-based decomposition suggests that the importance of genes in later skill differences operates increasingly through parental investments and family environments.

Main Results and Policy Interpretation

Skill formation

- ▶ Existing skills are persistent, but the estimated self-productivity is below one.
- ▶ Parental investment contributes to the accumulation of skills.
- ▶ Child and parental genes both matter for later skill development.
- ▶ The influence of genes increases over early childhood.

Investment function

- ▶ Parental genes have a strong association with parental investments.
- ▶ The direct effect of child genes on investment is smaller once current skills and parental genes are controlled for.
- ▶ A substantial part of the genetic association with skills operates through parental investments.

Equalizing investments can reduce, rather than increase, the importance of genetic differences in observed skills.

Application II: Human Capital Development in India

Attanasio, Meghir, and Nix

Motivation

Children from poor households often fall behind in both health and cognitive development early in life.

The paper estimates how health, cognition, parental background, and investments jointly shape child development.

- ▶ Child development is multidimensional.
- ▶ Health may affect later cognition.
- ▶ Parents may adjust investments in response to child outcomes.
- ▶ The timing of investment may be crucial.

The key setting is a developing country with large health deficits and substantial variation in household resources.

Research Questions

1. How persistent are cognitive skills and health?
2. Does early health affect later cognitive development?
3. How productive are parental investments at different ages?
4. Are cognition, health, parental background, and investment complementary?
5. How does poverty generate persistent inequality in human capital?

Methodological question

How can we estimate nonlinear production functions when both skills and investments are latent and parental investments are endogenous?

Data: Young Lives India

Longitudinal survey

- ▶ Younger cohort of the Young Lives Survey.
- ▶ Children are observed at approximately ages 1, 5, 8, and 12.
- ▶ Initial sample: approximately 2,011 children.
- ▶ Complete-round sample: approximately 1,910 children.
- ▶ The survey covers several districts in Andhra Pradesh and Telangana.

- ▶ The survey deliberately oversamples poor communities.
- ▶ It contains measures of health, cognition, parental background, investments, prices, and household resources.
- ▶ The sample is rich for studying mechanisms, but it is not nationally representative.

Two Dimensions of Human Capital

The paper treats child human capital as multidimensional:

$$\theta_t^C = \text{latent cognitive skill}, \quad \theta_t^H = \text{latent health}.$$

The two dimensions evolve jointly:

$$\theta_{t+1}^C = G_t \left(\theta_t^C, \theta_t^H, I_t, X_t \right),$$

$$\theta_{t+1}^H = F_t \left(\theta_t^C, \theta_t^H, I_t, X_t \right).$$

- ▶ Health can affect the production of cognition.
- ▶ Cognition can affect the production of health.
- ▶ Parental investment can affect both dimensions.
- ▶ Parental cognition and health enter as background factors.

Age-Specific CES Production Functions

For cognition, the production function is

$$\theta_{t+1}^C = A_t^C \left[\delta_{ct}^C (\theta_t^C)^{\rho_t} + \delta_{ht}^C (\theta_t^H)^{\rho_t} + \delta_{lt}^C (l_t)^{\rho_t} + \dots \right]^{1/\rho_t}.$$

For health,

$$\theta_{t+1}^H = A_t^H \left[\alpha_{ct}^H (\theta_t^C)^{\zeta_t} + \alpha_{ht}^H (\theta_t^H)^{\zeta_t} + \alpha_{lt}^H (l_t)^{\zeta_t} + \dots \right]^{1/\zeta_t}.$$

- ▶ The parameters vary by developmental stage.
- ▶ The production functions allow for interactions between cognition, health, and investment.
- ▶ The substitution parameters determine the degree of complementarity.
- ▶ The model imposes constant returns to scale within each production function.

A Reduced-Form Investment Function

The paper does not estimate a complete structural household model. Instead, it estimates a reduced-form investment equation:

$$\begin{aligned}\ln I_t = & \gamma_{0,t} + \gamma_{c,t} \ln \theta_t^C + \gamma_{h,t} \ln \theta_t^H \\ & + \gamma_{cp,t} \ln \theta_P^C + \gamma_{hp,t} \ln \theta_P^H + \gamma_{X,t} X_t \\ & + \gamma_{p,t} \ln p_t^I + \gamma_{Y,t} \ln Y_t + v_t.\end{aligned}$$

- ▶ p_t^I : prices of investment goods.
- ▶ Y_t : household resources.
- ▶ v_t : unobserved investment shock.
- ▶ Parents may respond to the child's current health and cognition.

The reduced-form specification avoids assuming that parents know the true production function.

Investment Endogeneity: The Control-Function Approach

Parents may adjust investment in response to unobserved shocks to the child's development.

Thus,

$$\text{Cov}(v_t, u_t^C) \neq 0, \quad \text{Cov}(v_t, u_t^H) \neq 0.$$

The paper uses a control-function approach:

$$E(u_t^C \mid Q_t, Z_t) = \kappa_C v_t,$$

$$E(u_t^H \mid Q_t, Z_t) = \kappa_H v_t.$$

1. Estimate the investment equation using prices and resources.
2. Obtain the residual $\hat{v}_t$.
3. Include $\hat{v}_t$ in the cognition and health production functions.

The Instruments for Investment

The excluded instruments are variables that affect investment but are assumed not to directly affect child development.

$$Z_t = [\text{local prices of investment goods}, \quad \text{household resources}].$$

- ▶ Prices affect the cost of food, clothing, medication, and educational goods.
- ▶ Household resources affect the ability to purchase investments.
- ▶ Prices are measured at the local level and are treated as supply-side variation.
- ▶ Resources are excluded from the production function conditional on family background and initial conditions.

Identification assumption

The instruments affect future cognition and health only through their effect on parental investment.

Measurement System and Age-Invariance

Each latent factor is measured by several noisy indicators:

$$m_{jkt} = a_{jkt} + \lambda_{jkt} \ln \theta_{kt} + e_{jkt}.$$

Latent factor	Main normalizing measure
Child cognition	PPVT
Child health	Height-for-age z-score
Parental investment	Expenditure on books
Parental health	Mother's weight
Parental cognition	Mother's years of schooling
Household resources	Household income

- ▶ Child cognition, health, and investment are allowed to grow over time.
- ▶ Only the initial mean is normalized to zero.
- ▶ The same anchor measure is used across periods to link the scale.
- ▶ This is an age-invariance restriction on the measurement mapping.

Three-Step Estimation Procedure

1. Estimate the distribution of observed measures

$$M = A + \Lambda \ln \theta + \Sigma e.$$

The observed measures are modeled as a mixture of normal distributions.

2. Recover latent-factor parameters

Minimum-distance restrictions recover:

$$\Lambda, \quad A, \quad \Sigma, \quad \text{Var}(\ln \theta).$$

These restrictions include factor normalizations and age-invariant anchor measures.

3. Estimate production and investment equations

Draw a synthetic data set from the estimated joint distribution and estimate the structural equations by regression.

MLE + minimum distance + simulated regression

Main Results: Health, Cognition, and Investment

Skill formation

- ▶ Early health has a substantial effect on later cognition.
- ▶ This effect is particularly important for children with poor initial health.
- ▶ Cognitive skills become persistent over childhood.
- ▶ Health affects cognition more strongly at younger ages.

Parental investment

- ▶ Investments affect cognitive development at all observed ages.
- ▶ The effect on cognition is larger at younger ages.
- ▶ Investments affect health mainly during the early stages of childhood.
- ▶ Ignoring investment endogeneity understates the effect of investment.

Early health interventions can generate cognitive gains through the dynamic interaction between health and cognition.

Policy Implications

The estimated model can be used to simulate interventions at different ages.

- ▶ Income transfers have their largest cognitive effect when provided during middle childhood.
- ▶ Income transfers have their largest health effect when provided earlier.
- ▶ Improving early health can generate persistent cognitive benefits.
- ▶ Later interventions remain valuable because cognitive development is not perfectly persistent.

Main takeaway

The optimal timing of intervention depends on the outcome:

Health intervention: earlier

Cognitive intervention: early and later investments both matter

References

- Francesco Agostinelli and Matthew Wiswall. Identification of dynamic latent factor models: The implications of re-normalization in a model of child development. NBER Working Paper 22441, National Bureau of Economic Research, 2016.
- Francesco Agostinelli and Matthew Wiswall. Estimating the technology of children's skill formation. *Journal of Political Economy*, 133(3):846–887, 2025. doi: 10.1086/733396. First circulated as NBER Working Paper 22442 (2016).
- Flavio Cunha and James J. Heckman. The technology of skill formation. *American Economic Review*, 97(2):31–47, 2007. doi: 10.1257/aer.97.2.31.
- Flavio Cunha, James J. Heckman, and Susanne M. Schennach. Estimating the technology of cognitive and noncognitive skill formation. *Econometrica*, 78(3): 883–931, 2010. doi: 10.3982/ECTA6551.